EXPLORATORY DATA ANALYSIS

Chapter 14

# Covariance and Dependence

Understanding joint variation beyond correlation

**Course:** From Raw Data to Understanding

**Focus:** Joint variation • Dependence • Nonlinear relationships • Multivariate reasoning

A mathematically grounded, practical chapter using Python, NumPy, Pandas, Matplotlib's object-oriented API, and Seaborn.

# Learning objectives

- Define covariance and understand its sign, units, and scale dependence.
- Derive the relationship between covariance, variance, and Pearson correlation.
- Distinguish zero covariance, zero correlation, and statistical independence.
- Recognize nonlinear and non-monotonic dependence that correlation can miss.
- Understand joint distributions, conditional distributions, and dependence structures.
- Use practical numerical and visual diagnostics to investigate multivariate dependence.
- Communicate the limits of dependence measures without making causal claims.

# How this chapter fits the course

Chapter 13 introduced Pearson, Spearman, and Kendall association measures, as well as categorical association. This chapter goes deeper into the mathematical idea of variables varying together. Covariance is foundational, but it is only one summary of joint behavior. Dependence is a much broader concept.

## Review of earlier chapters (excluding Chapter 13)

- Chapters 1–3: EDA is iterative; define the observational unit, inspect metadata and data quality, and understand missingness, duplicates, validity, and the data-generating process.
- Chapters 4–6: classify variables correctly; understand numerical, categorical, and temporal measurements, units, granularity, and temporal ordering.
- Chapters 7–9: use descriptive statistics, quantiles, distribution summaries, skewness, and kurtosis with attention to robustness and outliers.
- Chapters 10–11: inspect distributions and pairwise relationships visually; scatterplots can reveal curvature, clusters, outliers, and patterns hidden by a single coefficient.
- Chapter 12: use multivariate encodings, facets, heatmaps, and small multiples while making scales, aggregation, and denominators explicit.

## Review of the immediately previous chapter: Chapter 13 — Correlation and Association Measures

Pearson correlation summarizes linear association between numerical variables. Spearman and Kendall use ranks and are useful for monotonic or ordinal relationships. Categorical association can be investigated with contingency tables, chi-square tests, Phi, and Cramér's V. A coefficient is not a complete description: plots, sample size, effect size, missingness, outliers, and domain context remain important.

## Chapter 13 assignment/project: expected-reasoning guide

- A near-zero Pearson coefficient does not rule out a curved or otherwise nonlinear relationship.
- Spearman is often useful for ordinal or monotonic-but-nonlinear patterns; Kendall focuses on concordant and discordant pairs.
- Chi-square assesses compatibility with independence under its assumptions; Cramér's V describes the magnitude of categorical association.
- A small p-value is not the same as a practically important effect.
- An apparent pooled relationship may change after stratifying by a relevant variable.

This is an expected-reasoning guide, not a grading of submitted work.

## Preview of this chapter

We will start with covariance, derive its relationship to correlation, and then move beyond linear summaries. The key conceptual distinction is that independence is a statement about the entire joint distribution, while covariance captures only a particular form of joint variation.

# 1. Covariance: variables varying together

Covariance describes whether deviations of two variables from their respective means tend to have the same sign or opposite signs. When X is above its mean at the same observations where Y tends to be above its mean, covariance tends to be positive. When X is above its mean where Y tends to be below its mean, covariance tends to be negative.

**Population covariance:** $\mathrm{Cov}(X,Y) = E[(X - \mu X)(Y - \mu Y)]$

For n paired sample observations, the usual unbiased sample covariance estimator is:

**sXY = Σ[(xi – x■)(yi – ■)] / (n – 1)**

The denominator n – 1 is used for the conventional unbiased estimator when observations are sampled independently from a population with finite second moments. Dividing by n gives the empirical covariance of the observed dataset (often called the population-form covariance). The choice should be explicit when matching formulas or software outputs.

## 1.1 Sign and interpretation

- Positive covariance: paired deviations tend to point in the same direction.
- Negative covariance: paired deviations tend to point in opposite directions.
- Covariance near zero: little net linear co-variation in the observed scale; other dependence may remain.
- Covariance exactly zero: the cross-products sum or average to zero; it does not generally imply independence.

## 1.2 Units and magnitude

Covariance has the product of the variables' units. If X is measured in dollars and Y in kilograms, covariance is measured in dollar-kilograms. Its numerical magnitude is therefore hard to compare across variable pairs with different units or scales. This is one reason correlation is often easier to communicate.

## 1.3 A small numerical illustration

Suppose X = [1, 2, 3] and Y = [2, 4, 6]. Their means are 2 and 4. The products of centered deviations are 2, 0, and 2, whose sum is 4. The sample covariance is 4/(3–1) = 2. The positive value reflects that both variables rise together. If Y were [6, 4, 2], the centered products would sum to –4 and sample covariance would be –2.

# 2. Algebraic properties of covariance

Covariance has useful algebraic properties. These are not merely formal: they explain why units and transformations affect covariance.

- Symmetry: $\mathrm{Cov}(X,Y) = \mathrm{Cov}(Y,X)$.
- Covariance with a constant is zero: $\mathrm{Cov}(X,c) = 0$.
- Scaling: $\mathrm{Cov}(aX,bY) = ab\,\mathrm{Cov}(X,Y)$.
- Adding constants does not change covariance: $\mathrm{Cov}(X+a,Y+b) = \mathrm{Cov}(X,Y)$.
- Linearity: $\mathrm{Cov}(X+Z,Y) = \mathrm{Cov}(X,Y) + \mathrm{Cov}(Z,Y)$, when the required moments exist.
- Variance is covariance with itself: $\mathrm{Var}(X) = \mathrm{Cov}(X,X)$.

If one variable is multiplied by 10, covariance is multiplied by 10. If both are multiplied by 10, covariance is multiplied by 100. A negative multiplier reverses the sign. These facts make covariance meaningful mathematically but scale-dependent numerically.

# 3. From covariance to correlation

Pearson correlation standardizes covariance by dividing by the product of the standard deviations:

**ρXY = Cov(X,Y) / (σX σY)**

For a sample, r = sXY / (sX sY), provided both sample standard deviations are non-zero. The units cancel, producing a dimensionless measure bounded between –1 and +1.

### 3.1 Why is correlation bounded?

The Cauchy–Schwarz inequality implies that the absolute covariance cannot exceed the product of the standard deviations. Consequently, the absolute correlation cannot exceed 1. Equality occurs when one variable is an exact affine function of the other, apart from degenerate cases.

### 3.2 Covariance versus correlation

| Property | Covariance | Pearson correlation |
|---|---|---|
| Units | Product of variable units | None |
| Range | Unbounded | –1 to +1 |
| Scale changes | Magnitude changes | Unaffected by positive rescaling |
| Sign | Direction of linear co-variation | Direction of linear association |
| Comparison across pairs | Often difficult | More comparable, still contextual |

## 4. Independence, zero covariance, and zero correlation

**Independence** is stronger than uncorrelatedness. X and Y are independent if learning the value of one does not change the probability distribution of the other. For discrete variables, the joint probabilities factor into the product of marginal probabilities; for continuous variables, the analogous factorization applies to the joint density where defined.

> If X and Y are independent and their relevant expectations exist, then Cov(X,Y) = 0. The converse is not generally true. Zero covariance means no linear co-variation as measured by covariance; it does not rule out nonlinear dependence.

### 4.1 A classic counterexample

Let X be symmetrically distributed around zero and let $Y = X^2$. Large absolute values of X produce large Y, so Y clearly depends on X. Yet the covariance can be zero because positive and negative X values contribute opposite-signed cross-products that cancel.

For a simple symmetric example, let X take values –2, –1, 1, 2 with equal probability, and let $Y = X^2$. Then Y is completely determined by X, but Cov(X,Y) = 0.

### 4.2 Visual intuition

A scatterplot of $Y = X^2$ produces a U-shaped cloud. A straight-line summary can have slope near zero, even though the pattern is unmistakable. This is why a correlation matrix cannot replace plots and domain reasoning.

## 5. Dependence is a property of the joint distribution

The joint distribution describes how two or more variables behave together. Marginal distributions describe each variable separately. Two datasets can have the same marginal distributions but very different dependence structures. Looking only at individual histograms does not tell us how values are paired.

### 5.1 Conditional distributions

A useful way to investigate dependence is to compare the conditional distribution of Y at different values or ranges of X. If the distribution of Y changes with X, the variables are dependent. This change may involve the conditional mean, variance, shape, or tail behavior—not only a shift in average.

- Conditional mean changes: E[Y | X=x] varies with x.
- Conditional variance changes: spread in Y differs across values of X (heteroscedasticity).
- Conditional shape changes: skewness, modality, or tail behavior changes with X.
- Tail dependence: extreme values of one variable are more or less likely to coincide with extreme values of another.

### 5.2 Heteroscedasticity

Heteroscedasticity means that the conditional variance of one variable changes across levels of another variable or across fitted values. A funnel-shaped scatterplot is a familiar visual pattern. A near-zero correlation does not rule this out. It may matter for later statistical modeling, but in EDA the immediate task is to identify and describe the changing spread.

### 5.3 Tail dependence

Two variables may have little overall linear correlation but still tend to experience extreme values together. This can be important in risk, reliability, and operational settings. Tail behavior is not adequately summarized by one Pearson coefficient; inspect joint extremes and the domain-specific consequences.

## 6. Nonlinear and non-monotonic dependence

Dependence can take many forms. Pearson correlation focuses on linear co-variation; rank correlations focus on monotonic ordering. Neither captures every structure.

- Quadratic or U-shaped: Y changes with X but first rises and then falls, or vice versa.
- Periodic: Y oscillates as X changes, as in some seasonal or cyclic relationships.
- Threshold: Y remains stable until X crosses a meaningful boundary.
- Piecewise: the relationship has different regimes in different ranges.
- Variance-only: the conditional spread changes even when the conditional mean is stable.
- Mixture or subgroup: different populations follow different patterns, producing a complex pooled cloud.

A practical diagnostic sequence is to plot the raw pairs, inspect subgroup facets, consider a smooth trend only as a descriptive aid, and compare conditional distributions across useful ranges. Avoid forcing a complex relationship into a single coefficient.

## 7. Dependence measures beyond covariance and correlation

There is no universal dependence coefficient that is best for every dataset. Some measures are designed to detect broader forms of dependence, but they introduce their own assumptions, biases, and interpretation challenges.

### 7.1 Mutual information

Mutual information measures how much knowing one variable reduces uncertainty about another. For discrete variables it can be written as a sum over joint probabilities; for continuous variables it is defined using densities and may require estimation. It is zero under independence under standard conditions, and positive values indicate dependence.

Mutual information can detect nonlinear dependence that correlation misses. However, its estimated value depends on sample size, discretization or density-estimation choices, and estimator settings. It is not bounded by 1 and is less immediately interpretable than r.

### 7.2 Distance correlation

Distance correlation is based on pairwise distances and, under suitable finite-moment conditions, is zero if and only if the random vectors are independent. It can detect a wider range of dependence than ordinary correlation. Its sample estimate and inferential use still require care, especially with small samples or high dimensions.

### 7.3 Maximal information coefficient and other exploratory scores

Some exploratory dependence scores search over grids or function classes to detect varied patterns. Such methods can be useful for screening but may be computationally expensive, sensitive to tuning, and prone to false discoveries when many pairs are examined. Treat them as exploratory tools, not automatic truth detectors.

For this course, the main practical lesson is not to memorize every metric. It is to recognize the limits of covariance and correlation, then choose a method that matches the question.

## 8. Multivariate covariance and correlation matrices

For p numerical variables, the covariance matrix collects every pairwise covariance. It is a p × p symmetric matrix. Its diagonal contains the variances; off-diagonal entries contain pairwise covariances. The correlation matrix has 1s on the diagonal for non-constant variables and pairwise Pearson correlations off the diagonal.

A covariance matrix is positive semidefinite when constructed consistently from the same complete observations using the standard definition. Pairwise deletion with different samples for different entries can produce a matrix that is not positive semidefinite.

### 8.1 Reading the matrix

- Large positive covariance means variables tend to deviate from their means in the same direction, on their original scales.
- Large negative covariance means opposite-signed deviations are common.
- Correlation standardizes scale, but it still summarizes only linear pairwise association.
- Pairwise matrices do not describe all higher-order dependence among three or more variables.
- Missing-data policy and pairwise sample counts should be documented.

## 9. Python: covariance and correlation in practice

The example below compares linear and nonlinear relationships. Matplotlib figures are created with the object-oriented interface; Seaborn receives explicit Axes objects.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

rng = np.random.default_rng(42)
x = np.linspace(-3, 3, 250)
y_linear = 2.0 * x + rng.normal(0, 1.0, size=len(x))
y_curve = x ** 2 + rng.normal(0, 0.35, size=len(x))

data = pd.DataFrame({
    "x": x,
    "linear_response": y_linear,
    "curved_response": y_curve
})

# Sample covariance and correlation.
print(data[["x", "linear_response"]].cov())
print(data[["x", "linear_response"]].corr())

# Compare the linear and curved relationships visually.
fig, axes = plt.subplots(1, 2, figsize=(10, 4))
sns.scatterplot(data=data, x="x", y="linear_response",
                ax=axes[0], s=18, alpha=0.65)
sns.scatterplot(data=data, x="x", y="curved_response",
                ax=axes[1], s=18, alpha=0.65)
axes[0].set_title("Approximately linear dependence")
axes[1].set_title("Curved dependence")
fig.tight_layout()

print("Linear Pearson r:",
      data["x"].corr(data["linear_response"]))
print("Curved Pearson r:",
      data["x"].corr(data["curved_response"]))
```

> The curved example is designed so that positive and negative x values produce similar response values. Its Pearson correlation can therefore be close to zero even though the relationship is clearly not independent.

## 10. Conditional summaries as an EDA diagnostic

```
# Divide a numerical predictor into quantile-based bins.
data["x_band"] = pd.qcut(data["x"], q=6, duplicates="drop")

conditional = data.groupby("x_band", observed=True).agg(
    n=("curved_response", "size"),
    mean=("curved_response", "mean"),
    median=("curved_response", "median"),
    std=("curved_response", "std"),
    q25=("curved_response", lambda s: s.quantile(0.25)),
    q75=("curved_response", lambda s: s.quantile(0.75))
)
print(conditional)

# Plot conditional spread and central tendency.
fig, ax = plt.subplots(figsize=(8, 4.5))
centers = np.arange(len(conditional))
```

```
ax.errorbar(
    centers,
    conditional["mean"],
    yerr=conditional["std"],
    fmt="o",
    capsize=4
)
ax.set_xticks(centers)
ax.set_xticklabels(
    [str(interval) for interval in conditional.index],
    rotation=30, ha="right"
)
ax.set_ylabel("Curved response")
ax.set_title("Conditional mean and standard deviation by x-band")
fig.tight_layout()
```

Quantile bins are a convenient descriptive device, not a discovery of natural groups. The number of bins affects the picture. Check sample counts and avoid interpreting noisy small groups as stable patterns.

# 11. A practical dependence investigation workflow

- Define the question and observational unit; identify whether observations are independent, clustered, repeated, or time-ordered.
- Inspect each variable separately: range, shape, missingness, outliers, and measurement scale.
- Plot pairs using suitable encodings; add facets where meaningful subgroups may matter.
- Calculate covariance and Pearson correlation when linear co-variation is relevant.
- Compare rank correlations when monotonic or ordinal structure is relevant.
- Inspect conditional means, variances, quantiles, and distributions across predictor ranges.
- Investigate nonlinear, subgroup, and tail patterns where the domain suggests them.
- Record pairwise sample sizes, missing-data rules, units, and any transformations used.
- Report the observed structure and uncertainty without turning association into a causal claim.

# 12. Common mistakes

- Treating covariance magnitude as comparable across variables with different units.
- Assuming zero covariance or zero correlation means independence.
- Assuming a correlation matrix captures every kind of dependence.
- Looking only at conditional means while ignoring conditional variance or distribution shape.
- Using pairwise deletion without checking changing sample sizes or matrix consistency.
- Applying a complex dependence score without understanding its estimator and sample-size behavior.
- Interpreting a pooled relationship without checking meaningful subgroups.
- Treating a smooth curve as proof of a stable functional relationship.
- Confusing dependence with causation.

# 13. Summary

- Covariance measures linear joint variation and retains the product units of the variables.
- Correlation standardizes covariance, making it dimensionless and bounded between –1 and +1.
- Independence implies zero covariance when moments exist, but zero covariance does not generally imply independence.
- Dependence is a property of the joint distribution and can involve conditional means, variances, shapes, and tails.
- Pearson and rank correlations miss some nonlinear or non-monotonic structures.
- Conditional summaries, plots, and domain reasoning complement coefficients.
- Advanced measures such as mutual information and distance correlation can detect broader dependence, but require careful estimation and interpretation.

# 14. Assignment

## Part A — Concepts

- Explain why covariance changes when the measurement unit of one variable changes, while Pearson correlation does not change under positive rescaling.
- Give a concrete example of dependent variables with zero covariance. Explain the cancellation.
- State the implication that independence has for covariance, and explain why the reverse implication fails.
- Describe a relationship in which conditional variance changes while the conditional mean remains approximately constant.
- Why can two datasets have identical marginal distributions but different dependence structures?

## Part B — Calculation

Let X = [1, 2, 3, 4] and Y = [8, 6, 4, 2]. Calculate the sample means, sample covariance (using n – 1), sample standard deviations, and Pearson correlation. Interpret the sign and magnitude. Then multiply X by 100 and explain what changes and what remains invariant.

## Part C — Investigation

Generate a symmetric X variable and define $Y = X^2$ plus a small amount of random noise. Calculate covariance, Pearson correlation, and Spearman correlation. Create a scatterplot. Explain why a low Pearson coefficient does not establish independence.

## Part D — Conditional structure

Use a dataset with at least two numerical variables. Divide one variable into meaningful ranges or quantile bins, then compare the conditional mean, standard deviation, and quartiles of the other variable. Describe any changing spread or shape.

# 15. Coding project — Dependence Investigation Notebook

**Goal:** Extend the cumulative EDA master notebook with a section that investigates joint variation beyond a standard correlation matrix.

- Choose a dataset with at least three numerical variables and a meaningful observational unit.
- Document units, sampling structure, repeated observations, and missingness.
- Calculate and interpret a covariance matrix and a Pearson correlation matrix.
- Compare at least two pairs where covariance and correlation tell different things about scale and comparability.
- Find or construct a pair with a nonlinear pattern that a linear coefficient summarizes poorly.
- Use plots and conditional summaries to investigate means, spread, and quantiles across ranges.
- Where relevant, compare pooled patterns with subgroup patterns.
- Write a short report distinguishing linear co-variation, broader dependence, and causation.
- Include limitations, pairwise sample sizes, and reproducible code.

**Deliverables:** a well-organized notebook section, figures, summary tables, and a concise interpretation. Keep valid observations unless a documented data-quality reason justifies exclusion.

# Preview of Next Chapter

**Chapter 15 — Grouped Analysis: GroupBy, Crosstabs, and Pivot Tables**

We will learn to compare distributions and summaries across groups, construct and normalize contingency tables, and use Pandas groupby and pivot_table workflows. The emphasis will be on choosing denominators, preserving observational meaning, and avoiding misleading aggregate comparisons.

# Selected references and further reading


- Freedman, D., Pisani, R., & Purves, R. Statistics. Sections on covariance, correlation, and independence.
- Casella, G., & Berger, R. L. Statistical Inference. Foundational treatment of random variables and dependence.
- Cover, T. M., & Thomas, J. A. Elements of Information Theory. Mutual information and dependence.
- SciPy documentation: scipy.stats and distance-based statistical tools.
- Pandas documentation: DataFrame.cov, DataFrame.corr, groupby, and qcut.
- Seaborn documentation: relational plots and distribution plots.


This chapter is educational material for exploratory data analysis. Dependence measures describe statistical structure; causal conclusions require an appropriate study design and additional assumptions.